

Substantive Citation Report

Twenty-one substantive citing articles, each published in an ICORE2026 A*-rated conference, cited more than 100 times on Google Scholar, or using or extending TQT in its own methods or experiments. Articles are ordered by venue standing and citations, the citing work's reliance on TQT and the credit it gives TQT, taken together.

21

selected articles

100

marked discussion pages

30

countries / territories in
the wider survey

Selection and source documentation

The selection basis lists every criterion an article meets. ICORE2026 A* venue: published in a conference rated A* in ICORE2026; ACM Transactions on Graphics 39(6) contains the SIGGRAPH Asia 2020 technical papers, and CVPR workshops carry no separate rating. >100 Scholar citations: more than 100 Google Scholar citations in the saved snapshot. Uses or extends TQT: the article's own methods or experiments use, follow or extend TQT, or compare an implemented extension against it, as stated in the source passages; background mentions, comparisons without such use and statements that a design is merely similar to TQT do not qualify. Counts belong to the citing articles' Scholar records or version clusters; two records show no cited-by count in the saved snapshot. Each entry identifies its reviewed publication version, the citing authors' affiliations, any employer link and the contribution supported by the source passage.

Authorship links and documented scholarly use

Each article identifies the cited work and authors, the specific research connection and the original discussion pages. The bibliography link establishes authorship of the corresponding work by Sambhav R. Jain and coauthors.

Orange rectangles mark selected discussion, citations and TQT results; blue rectangles identify the corresponding bibliography entries. The marked packet follows the article order in this report and retains original page numbers. Article numbers identify the corresponding marked files.

International citation survey. The wider survey maps 210 citing works to 228 institution/location entries across 30 countries/territories. These totals describe the broader citation network, separately from this report's selection.

[Report PDF](#) · [Marked discussions](#) · [Interactive map](#)

Snapshot: October 7, 2026. [Sambhav Jain's Google Scholar profile](#). Counts belong to the saved citing-article records or version clusters.

Selected citing articles (21)

No.	Article / venue	Scholar	Selection basis
01	Mixed Precision DNNs: All you need is a good parametrization ICLR 2020	230	ICORE2026 A* venue >100 Scholar citations Uses or extends TQT
02	Understanding How Orthogonality of Parameters Improves Quantization of Neural Networks IEEE Transactions on Neural Networks and Learning Systems 34(12), 10737-10746 (2023)	12	Uses or extends TQT
03	Bayesian Bits: Unifying Quantization and Pruning NeurIPS 2020	207	ICORE2026 A* venue >100 Scholar citations
04	A Reduced-Precision Network for Image Reconstruction ACM Transactions on Graphics 39(6), Article 231 (SIGGRAPH Asia 2020)	41	ICORE2026 A* venue Uses or extends TQT
05	7 μJ/inference end-to-end gesture recognition from dynamic vision sensor data using ternarized hybrid convolutional neural networks Future Generation Computer Systems 149 (2023), 717-731	12	Uses or extends TQT
06	Understanding and Overcoming the Challenges of Efficient Transformer Quantization EMNLP 2021	271	ICORE2026 A* venue >100 Scholar citations Uses or extends TQT
07	F8Net: Fixed-Point 8-bit Only Multiplication for Network Quantization ICLR 2022	72	ICORE2026 A* venue
08	Improving Low-Precision Network Quantization via Bin Regularization ICCV 2021, 5241-5250	65	ICORE2026 A* venue Uses or extends TQT
09	A Resource Efficient Integer-Arithmetic-Only FPGA-Based CNN Accelerator for Real-Time Facial Emotion Recognition IEEE Access 9 (2021), 104367-104381	54	Uses or extends TQT
10	Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy Electronics 10(22), 2823 (2021)	15	Uses or extends TQT
11	Optimizing the Deployment of Tiny Transformers on Low-Power MCUs IEEE Transactions on Computers 74(2), 526-541 (2025)	47	Uses or extends TQT

No.	Article / venue	Scholar	Selection basis
12	Accelerating configuration of Reconfigurable Intelligent Surfaces through a hardware-enhanced deep learning approach Computers and Electrical Engineering 134 (2026), 111101	None shown	Uses or extends TQT
13	Up or Down? Adaptive Rounding for Post-Training Quantization ICML 2020	1,224	ICORE2026 A* venue >100 Scholar citations
14	Bringing AI to edge: From deep learning's perspective Neurocomputing 485 (2022)	247	>100 Scholar citations
15	HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs ECCV 2020	95	ICORE2026 A* venue
16	A Low Memory Footprint Quantized Neural Network for Depth Completion of Very Sparse Time-of-Flight Depth Maps CVPR Workshops 2022 (CVPRW), 2686–2695	12	Uses or extends TQT
17	QKD: Quantization-aware Knowledge Distillation arXiv preprint, 2019	125	>100 Scholar citations Uses or extends TQT
18	FELIX: A Ferroelectric FET Based Low Power Mixed-Signal In-Memory Architecture for DNN Acceleration ACM Transactions on Embedded Computing Systems 21(6), Article 84 (2022)	50	Uses or extends TQT
19	A White Paper on Neural Network Quantization arXiv / Qualcomm white paper, 2021	1,362	>100 Scholar citations
20	INT8 Activation Ternary or Binary Weights Networks: Unifying Between INT8 and Lower-bit Width Quantization Journal of Robotics, Networking and Artificial Life 9(2) (2022), 171–176	None shown	Uses or extends TQT
21	Efficient SAR Vessel Detection for FPGA-Based On-Satellite Sensing SEC '25, Tenth ACM/IEEE Symposium on Edge Computing (2025)	6	Uses or extends TQT

SELECTED CITING ARTICLES

ARTICLE 01 · SUBSTANTIVE CITATION REPORT

Mixed Precision DNNs: All you need is a good parametrization

Stefan Uhlich; Lukas Mauch; Fabien Cardinaux; Kazuki Yoshiyama; Javier Alonso García; Stephen Tiedemann; Thomas Kemp; Akira Nakamura

ICLR 2020 · 230 Google Scholar citations

ICORE2026 A* venue

>100 Scholar citations

Uses or extends TQT

Documented contribution: Implemented experimental comparison

Sony researchers describe how TQT uses the straight-through estimator to learn the quantizer's dynamic range by gradient descent (p. 2) and cite it for the power-of-two step size that efficient hardware requires (p. 4). They train a TQT model, learning $q_{\max}$ at a fixed bitwidth "as was done in (Jain et al., 2019)", as the fixed-bitwidth benchmark for their concurrently developed mixed-precision method, and report the TQT rows in their CIFAR-10 and ImageNet result tables (pp. 8–9).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10](#).

Reviewed version: Author-hosted arXiv copy bearing ICLR 2020 publication header. **Pinpoint:** PDF/printed pp. 1–2, 4–5, 8–9, introduction, quantizer constraints, experimental setup and TQT result rows; bibliography p. 10. Source E-0087; marked copy E-1439.

Count: E-0430, `data.organic_results[11].inline_links.cited_by.total`. Venue: E-0328.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 01-Mixed-Precision-DNNs-MARKED.pdf

ARTICLE 02 · SUBSTANTIVE CITATION REPORT

Understanding How Orthogonality of Parameters Improves Quantization of Neural Networks

Sukru Burc Eryilmaz; Aysegul Dunder

IEEE Transactions on Neural Networks and Learning Systems 34(12), 10737–10746 (2023) · 12 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological adoption

NVIDIA and Bilkent University researchers adopt TQT as the baseline quantized retraining method for their study of orthogonality regularization, describing it as achieving state-of-the-art results (pp. 1–2). They state "we use TQT [25] as our baseline retraining" (p. 5), train thresholds in the log domain as described in [25] (p. 4), follow its five-epoch retraining schedule (p. 5) and combine the orthogonality penalty with TQT, reporting results with and without TQT in Section V-A (p. 6).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 10](#).

Reviewed version: Final IEEE TNNLS print, vol. 34, no. 12; public excerpt reproduces the title, discussion and bibliography pages. **Pinpoint:** Physical PDF pp. 1–6 (printed pp. 10737–10742): TQT introduced as the quantized retraining method, log-domain threshold training as described in [25], TQT baseline retraining and experiments with TQT; bibliography p. 10 (printed p. 10746), reference [25]. Source E-1436; marked copy E-1440.

Count: E-0402, `data.organic_results[19].inline_links.cited_by.total`. Venue: E-1429.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 02-Orthogonality-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 03 · SUBSTANTIVE CITATION REPORT

Bayesian Bits: Unifying Quantization and Pruning

Mart van Baalen; Christos Louizos; Markus Nagel; Rana Ali Amjad; Ying Wang; Tijmen Blankevoort; Max Welling
NeurIPS 2020 · 207 Google Scholar citations

ICORE2026 A* venue

>100 Scholar citations

Documented contribution: Attribution of a research contribution

The Bayesian Bits authors credit two papers, LSQ [8] and reference [15], the TQT work by Jain, Gural, Wu and Dick, with independently introducing joint learning of the quantization scale and network parameters for a fixed-bitwidth network (p. 6). Figure 2b then compares Bayesian Bits on MobileNetV2 with AdaRound, LSQ and TQT (p. 8).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10](#).

Reviewed version: Published NeurIPS PDF. **Pinpoint:** PDF/printed pp. 6, 8, scale learning and TQT in Figure 2b and its caption; bibliography p. 10, reference [15]. Source E-0324; marked copy E-1441.

Count: E-0430, data.organic_results[9].inline_links.cited_by.total. Venue: E-0331.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 03-Bayesian-Bits-MARKED.pdf

ARTICLE 04 · SUBSTANTIVE CITATION REPORT

A Reduced-Precision Network for Image Reconstruction

Manu Mathew Thomas; Karthik Vaidyanathan; Gabor Liktó; Angus G. Forbes
ACM Transactions on Graphics 39(6), Article 231 (SIGGRAPH Asia 2020) · 41 Google Scholar citations

ICORE2026 A* venue

Uses or extends TQT

Documented contribution: Direct methodological adoption

University of California, Santa Cruz and Intel researchers explain that Jain et al. and Esser et al. showed quantization thresholds could be trained, with a straight-through-estimator gradient that lets thresholds grow for dynamic range or shrink for precision (p. 3). For the quantization of their 4-bit QW-Net image-reconstruction network they state "Following Jain et al. [2019], we train the thresholds in log space" to improve stability during quantized training (Section 5.1, p. 6).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 11](#).

Reviewed version: ACM Transactions on Graphics publisher PDF. **Pinpoint:** Physical PDF pp. 3 and 6 (Section 5.1); trained quantization thresholds and log-space threshold training following Jain et al. [2019]; bibliography p. 11. Source E-1263; marked copy E-1442.

Count: E-0379, data.organic_results[0].inline_links.cited_by.total. Venue: E-1459; E-1460.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 04-QW-Net-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 05 · SUBSTANTIVE CITATION REPORT

7 μ J/inference end-to-end gesture recognition from dynamic vision sensor data using ternarized hybrid convolutional neural networks

Georg Rutishauser; Moritz Scherer; Tim Fischer; Luca Benini

Future Generation Computer Systems 149 (2023), 717–731 · 12 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological use and comparison

ETH Zürich and University of Bologna researchers train their ternarized gesture-recognition networks with two quantization-aware training algorithms, INQ and TQT. They describe TQT as optimizing the logarithm of the clipping interval (p. 5), use it to learn both weight and activation clipping bounds (p. 6), list its hyperparameters (p. 9) and compare it with INQ in Table 4, where ReLU activations with TQT give the highest classification accuracy (p. 10).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 14](#).

Reviewed version: Published open-access journal PDF (Elsevier). **Pinpoint:** Physical PDF pp. 5–6, 9–10: TQT as one of two quantization-aware training algorithms, TQT hyperparameters and Table 4 TQT versus INQ results; bibliography p. 14, reference [67]. Source E-1254; marked copy E-1443.

Count: E-0382, data.organic_results[9].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 05-DVS-Gesture-MARKED.pdf

ARTICLE 06 · SUBSTANTIVE CITATION REPORT

Understanding and Overcoming the Challenges of Efficient Transformer Quantization

Yelysei Bondarenko; Markus Nagel; Tijmen Blankevoort

EMNLP 2021 · 271 Google Scholar citations

ICORE2026 A* venue

>100 Scholar citations

Uses or extends TQT

Documented contribution: Methodological adaptation

Qualcomm researchers note that weight and activation ranges can be learned jointly with the weights during training, as in Esser et al. and Jain et al. (p. 3), and adapt this jointly credited procedure to learn weight and activation ranges for BERT-like transformer models in their quantization-aware training variant (p. 7).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 11](#).

Reviewed version: Published proceedings PDF. **Pinpoint:** Physical PDF pp. 3, 7, learnable weight/activation ranges and their adaptation to transformers; bibliography p. 11 / printed p. 7957. Source E-0046; marked copy E-1444.

Count: E-0430, data.organic_results[6].inline_links.cited_by.total. Venue: E-0326.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 06-Transformer-Quantization-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 07 · SUBSTANTIVE CITATION REPORT

F8Net: Fixed-Point 8-bit Only Multiplication for Network Quantization

Qing Jin; Jian Ren; Richard Zhuang; Sumant Hanumante; Zhengang Li; Zhiyu Chen; Yanzhi Wang; Kaiyuan Yang; Sergey Tulyakov

ICLR 2022 · 72 Google Scholar citations

ICORE2026 A* venue

Documented contribution: Comparison and design positioning

Snap Inc., Northeastern University and Rice University researchers position F8Net against the fixed-point quantization of Jain et al. (2019), which uses trainable power-of-two scale factors so that multiplication becomes bit shifting, and state that their work differs from it in three major aspects (pp. 2–3). They contrast F8Net's on-the-fly fractional length with TQT's directly trained fractional length (p. 7) and list TQT (Jain et al., 2019) in Table 1 for MobileNet V1 and V2b as the previous fixed-point quantization method (p. 8).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 12](#).

Reviewed version: arXiv:2202.05239 PDF carrying the ICLR 2022 conference header. **Pinpoint:** Physical PDF pp. 2–3, 7–8: positioning against the power-of-two fixed-point scaling of Jain et al. (2019), contrast with directly trained fractional lengths and TQT rows in Table 1; bibliography p. 12. Source E-0725; marked copy E-1445.

Count: E-0381, data.organic_results[1].inline_links.cited_by.total. Venue: E-0328.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 07-F8Net-MARKED.pdf

ARTICLE 08 · SUBSTANTIVE CITATION REPORT

Improving Low-Precision Network Quantization via Bin Regularization

Tiantian Han; Dong Li; Ji Liu; Lu Tian; Yi Shan

ICCV 2021, 5241–5250 · 65 Google Scholar citations

ICORE2026 A* venue

Uses or extends TQT

Documented contribution: Extension of the TQT baseline

Xilinx researchers in Beijing re-implement a hardware-friendly quantization baseline following references [19, 21], where [19] is TQT: power-of-two step sizes, low-bit first and last layers, quantized inputs and outputs of element-wise additions, 8-bit biases and batch normalization merged during training. They build bin regularization on this TQT baseline; Table 9 reports 63.8% top-1 for TQT, 65.8% for TQT+L2 and 67.5% for TQT+BR on 4-bit MobileNetV2, which the text describes as +3.7% over TQT (p. 8). Employer-linked: the authors list Xilinx Inc.

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 9](#).

Reviewed version: IEEE Xplore published print, ICCV 2021 proceedings; public excerpt reproduces the title, discussion and bibliography pages. **Pinpoint:** Physical PDF p. 8 (printed p. 5248): Section 4.8 hardware-friendly baseline following [19, 21], Table 9 TQT, TQT+L2 and TQT+BR rows and the reported +3.7% over TQT; bibliography p. 9 (printed p. 5249), reference [19]. Source E-1437; marked copy E-1446.

Count: E-0391, data.organic_results[7].inline_links.cited_by.total. Venue: E-1396.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 08-Bin-Regularization-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 09 · SUBSTANTIVE CITATION REPORT

A Resource Efficient Integer-Arithmetic-Only FPGA-Based CNN Accelerator for Real-Time Facial Emotion Recognition

Jaemyung Kim; Jin-Ku Kang; Yongwoo Kim

IEEE Access 9 (2021), 104367–104381 · 54 Google Scholar citations

Uses or extends TQT

Documented contribution: Method extension and implemented comparison

Inha University and Sangmyung University researchers build log level threshold quantization (LLTQ) on TQT's power-of-two scale mapping, where a trainable log-threshold turns scale multiplication into shifts (p. 6). LLTQ maps the trainable parameter directly to a power-of-two scale and initializes activation scales from first-batch statistics in place of TQT's pre-calibration. They illustrate LSQ, TQT and LLTQ (Figure 3, p. 7), train all three (p. 10) and report TQT as hardware-friendly within about 0.45% of floating-point accuracy, a gap LLTQ closes (p. 11).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 15](#).

Reviewed version: IEEE Access open-access publisher PDF. **Pinpoint:** Physical PDF pp. 3, 6–7, 10–11: TQT in related work, LLTQ derived as an alternative to TQT's power-of-two scale mapping, Figure 3 and the implemented LSQ/TQT/LLTQ comparison; bibliography p. 15, reference [39]. Source E-1264; marked copy E-1447.

Count: E-0393, data.organic_results[0].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 09-LLTQ-MARKED.pdf

ARTICLE 10 · SUBSTANTIVE CITATION REPORT

Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy

Maarten Vandersteegen; Kristof Van Beeck; Toon Goedemé

Electronics 10(22), 2823 (2021) · 15 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological adoption

KU Leuven researchers adopt TQT's quantizer: they optimize power-of-two scales by backpropagation "as proposed by Jain et al." (p. 12) and quantize activations with its symmetric uniform quantizer, hyperparameters and scale freezing (p. 17). They credit Jain et al. with showing that 8-bit models with power-of-two scales reach full-precision accuracy (p. 14), discuss TQT's shared elementwise-addition quantizer parameters (p. 12), extend its gradual freezing and stabilization criteria to per-channel scales (p. 19) and report TQT on ImageNet (Table 2, p. 20).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 23](#).

Reviewed version: Published journal PDF from KU Leuven repository. **Pinpoint:** PDF/printed pp. 11–14, 17–20, quantizer design, adopted training practices and TQT Table 2 results; bibliography p. 23, reference [12]. Source E-0081; marked copy E-1448.

Count: E-0459, data.organic_results[0].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 10-Integer-Only-CNNs-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 11 · SUBSTANTIVE CITATION REPORT

Optimizing the Deployment of Tiny Transformers on Low-Power MCUs

Victor J.B. Jung; Alessio Burrello; Moritz Scherer; Francesco Conti; Luca Benini

IEEE Transactions on Computers 74(2), 526–541 (2025) · 47 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological use

Researchers in the ETH Zürich/University of Bologna PULP group quantize their tiny transformers for low-power MCUs with TQT, stating "We use the uniform symmetric quantizer from the SotA Trained Quantization Threshold (TQT) method [3]", and specify the functional behavior of the TQT quantizer used in their deployment flow (p. 6).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 15](#).

Reviewed version: Published IEEE Transactions on Computers print; public excerpt reproduces the title, discussion and bibliography pages. **Pinpoint:** Physical PDF p. 6 (printed p. 531), Section IV-C: the uniform symmetric TQT quantizer used in the QuantLib flow; bibliography p. 15 (printed p. 540), reference [3]. Source E-1434; marked copy E-1449.

Count: E-0375, data.organic_results[6].inline_links.cited_by.total. Venue: E-1421.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 11-Tiny-Transformers-MARKED.pdf

ARTICLE 12 · SUBSTANTIVE CITATION REPORT

Accelerating configuration of Reconfigurable Intelligent Surfaces through a hardware-enhanced deep learning approach

Rubén Padial-Allué; Alberto Martín-Martín; Antonio García; Encarnación Castillo; José David Fernández-Rodríguez; Marcos Baena-Molina; Luis Parrilla

Computers and Electrical Engineering 134 (2026), 111101 · No cited-by count in the saved Google Scholar record

Uses or extends TQT

Documented contribution: Direct methodological use

University of Granada-led researchers use TQT in the quantization-aware training of their AMD DPU implementation for configuring reconfigurable intelligent surfaces, stating "For the quantization strategy, a power-of-two scale quantizer with training quantization threshold (TQT) was used [50]" (Section 3.2, p. 9). TQT is applied together with activation and training refinements, so the reported results reflect the combined pipeline.

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 17](#).

Reviewed version: Open-access Elsevier PDF (CC BY), University of Granada repository copy. **Pinpoint:** Physical PDF p. 9, Section 3.2: power-of-two scale quantizer with TQT for the AMD DPU implementation; bibliography p. 17, reference [50]. Source E-1185; marked copy E-1450.

Count: E-0385, data.organic_results[1].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 12-RIS-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 13 · SUBSTANTIVE CITATION REPORT

Up or Down? Adaptive Rounding for Post-Training Quantization

Markus Nagel; Rana Ali Amjad; Mart van Baalen; Christos Louizos; Tijmen Blankevoort

ICML 2020 · 1,224 Google Scholar citations

ICORE2026 A* venue

>100 Scholar citations

Documented contribution: Named technical recognition

In the related-work analysis of AdaRound, Qualcomm researchers name Jain et al., jointly with Esser et al., for learning the quantization minimum and maximum ranges during training so that they do not have to be set manually (p. 5).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10](#).

Reviewed version: Published PMLR PDF; physical pages retained. **Pinpoint:** Physical PDF p. 5, section 4 Background and related work; bibliography physical p. 10. Source E-0321; marked copy E-1451.

Count: E-0430, data.organic_results[2].inline_links.cited_by.total. Venue: E-0329.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 13-AdaRound-MARKED.pdf

ARTICLE 14 · SUBSTANTIVE CITATION REPORT

Bringing AI to edge: From deep learning's perspective

Di Liu; Hao Kong; Xiangzhong Luo; Weichen Liu; Ravi Subramaniam

Neurocomputing 485 (2022) · 247 Google Scholar citations

>100 Scholar citations

Documented contribution: Named survey discussion

The edge-AI survey describes Jain et al.'s trained uniform quantization method for accurate and efficient neural-network inference on fixed-point hardware (p. 9) and lists TQT as an 8-bit technique for weights and activations in its comparison table (p. 8); reference [121] identifies Jain et al.'s paper.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 20](#).

Reviewed version: arXiv author manuscript; journal publication verified separately. **Pinpoint:** PDF/printed pp. 8–9, TQT comparison-table row and named quantization discussion; bibliography p. 20, reference [121]; affiliations p. 1. Source E-0044; marked copy E-1452.

Count: E-0366, data.organic_results[0].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 14-Bringing-AI-to-Edge-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 15 · SUBSTANTIVE CITATION REPORT

HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs

Hai Victor Habi; Roy H. Jennings; Arnon Netzer

ECCV 2020 · 95 Google Scholar citations

ICORE2026 A* venue

Documented contribution: Quantizer-design connection

Sony Semiconductor Israel researchers cite references [24, 41] for hardware-friendly quantizers that are per-tensor, uniform, symmetric and use power-of-two thresholds (p. 2), call these properties an imperative requirement for efficient edge hardware "(see [24])", and state that the quantizer used in HMQ is similar to the one in [24], the TQT work, before defining its signed and unsigned forms in equations 2 and 3 (p. 4).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 17](#).

Reviewed version: Author arXiv manuscript; ECCV publication verified separately. **Pinpoint:** Physical PDF pp. 1–4, hardware-friendly quantization citations and quantizer definition; bibliography p. 17. Source E-0043; marked copy E-1453.

Count: E-0368, `data.organic_results[0].inline_links.cited_by.total`. Venue: E-0585.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 15-HMQ-MARKED.pdf

ARTICLE 16 · SUBSTANTIVE CITATION REPORT

A Low Memory Footprint Quantized Neural Network for Depth Completion of Very Sparse Time-of-Flight Depth Maps

Xiaowen Jiang; Valerio Cambarelli; Gianluca Agresti; Cynthia Ifeyinwa Ugwu; Adriano Simonetto; Fabien Cardinaux; Pietro Zanuttigh

CVPR Workshops 2022 (CVPRW), 2686–2695 · 12 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological use

EPFL, Sony and University of Padova researchers report that their most effective uniform-precision quantization-aware training procedure initializes from the pretrained float32 model and uses learnable qmax as in the Trained Quantization Threshold (TQT) approach [23], stabilized with cosine learning-rate decay and RMSprop followed by Adam (p. 5).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 9](#).

Reviewed version: IEEE Xplore published print, CVPRW 2022 proceedings; public excerpt reproduces the title, discussion and bibliography pages. **Pinpoint:** Physical PDF p. 5, Section 5.1 (printed p. 2690): uniform-precision QAT using learnable qmax as in TQT [23]; bibliography p. 9 (printed p. 2694), reference [23]. Source E-1435; marked copy E-1454.

Count: E-0402, `data.organic_results[14].inline_links.cited_by.total`. Venue: E-1422.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 16-Depth-Completion-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 17 · SUBSTANTIVE CITATION REPORT

QKD: Quantization-aware Knowledge Distillation

Jangho Kim; Yash Bhalgat; Jinwon Lee; Chirag Patel; Nojun Kwak
arXiv preprint, 2019 · 125 Google Scholar citations

>100 Scholar citations

Uses or extends TQT

Documented contribution: Use in a research implementation

Seoul National University and Qualcomm researchers describe LSQ [8] and TQT [17] as introducing uniform quantization with trainable interval values and state that QKD leverages these trainable approaches in its baseline quantization implementation (p. 2), with trainable interval parameters for each layer's weights and input activations similar to [8, 17, 45] (p. 3). They also cite TQT on the quantization sensitivity of depthwise convolutions (p. 7). Knowledge distillation is applied on top of this quantization baseline, and the reported results cover the combined method.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 9.](#)

Reviewed version: Reviewed arXiv manuscript. **Pinpoint:** Physical PDF pp. 2-3, 7, trainable intervals, dequantization and depthwise-layer sensitivity; bibliography p. 9, reference [17]. Source E-0056; marked copy E-1455.

Count: E-0378, data.organic_results[0].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 17-QKD-MARKED.pdf

ARTICLE 18 · SUBSTANTIVE CITATION REPORT

FELIX: A Ferroelectric FET Based Low Power Mixed-Signal In-Memory Architecture for DNN Acceleration

Taha Soliman; Nellie Laleni; Tobias Kirchner; Franz Müller; Ashish Shrivastava; Thomas Kämpfe; Andre Guntoro; Norbert Wehn

ACM Transactions on Embedded Computing Systems 21(6), Article 84 (2022) · 50 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological use

Robert Bosch GmbH, Fraunhofer IPMS and University of Kaiserslautern researchers evaluate the classification accuracy of their ferroelectric-FET in-memory accelerator with 8-bit/4-bit (input/weight) quantization "based on the optimizations presented in [17]", the TQT paper (Section 6.2, p. 17).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 23.](#)

Reviewed version: ACM Transactions on Embedded Computing Systems publisher PDF. **Pinpoint:** Physical PDF p. 17, Section 6.2: 8-bit/4-bit quantization based on the optimizations in [17]; bibliography p. 23, reference [17]. Source E-1279; marked copy E-1401.

Count: E-0381, data.organic_results[4].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 18-FELIX-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 19 · SUBSTANTIVE CITATION REPORT

A White Paper on Neural Network Quantization

Markus Nagel; Marios Fournarakis; Rana Ali Amjad; Yelysei Bondarenko; Mart van Baalen; Tijmen Blankevoort
arXiv / Qualcomm white paper, 2021 · 1,362 Google Scholar citations

>100 Scholar citations

Documented contribution: Technical exposition

Qualcomm's white paper on neural network quantization credits Jain et al., with Esser et al. and Bhalgat et al., for using the straight-through estimator to compute gradients with respect to the quantization parameters, then develops the gradient derivation for the scale factor (p. 20). It also cites Jain et al. among works that start quantization-aware training from a pretrained FP32 model (p. 22).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 26.](#)

Reviewed version: Reviewed arXiv white paper. **Pinpoint:** Physical PDF pp. 20, 22, gradient derivation and QAT initialization; bibliography p. 26. Source E-0090; marked copy E-1456.

Count: E-0430, data.organic_results[1].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 19-White-Paper-MARKED.pdf

ARTICLE 20 · SUBSTANTIVE CITATION REPORT

INT8 Activation Ternary or Binary Weights Networks: Unifying Between INT8 and Lower-bit Width Quantization

Ninnart Fuengfusin; Hakaru Tamukoh

Journal of Robotics, Networking and Artificial Life 9(2) (2022), 171–176 · No cited-by count in the saved Google Scholar record

Uses or extends TQT

Documented contribution: Direct methodological use

Kyushu Institute of Technology researchers build ITBWN by quantizing activations to INT8 with TQT while using binary or ternary weights, describing TQT's power-of-two scale, removal of the zero point and threshold training with the straight-through estimator (p. 2). TQT quantizes the activations in their binary, ternary and INT8 configurations (pp. 3–4) through the Xilinx Brevitas implementation, whose default settings, as the authors note, omit the power-of-two scale constraint and initialize the scale differently (p. 3).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 5.](#)

Reviewed version: Published journal PDF (J-STAGE). **Pinpoint:** Physical PDF pp. 2–5: TQT as the INT8 method, activations quantized with TQT via Xilinx Brevitas, TQT-based settings and conclusion; bibliography p. 5, reference 11. Source E-0595; marked copy E-1457.

Count: E-0422, data.organic_results[9].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 20-ITBWN-MARKED.pdf

SELECTED CITING ARTICLES

ARTICLE 21 · SUBSTANTIVE CITATION REPORT

Efficient SAR Vessel Detection for FPGA-Based On-Satellite Sensing

Colin Laganier; Liam Fletcher; Elim Kwan; Richard Walters; Victoria Nockles

SEC '25, Tenth ACM/IEEE Symposium on Edge Computing (2025) · 6 Google Scholar citations

Uses or extends TQT

Documented contribution: Methodological use via Vitis AI

The Alan Turing Institute researchers deploy a YOLOv8-based SAR vessel detector on an AMD/Xilinx Kria KV260 FPGA, quantizing it to INT8 with the Vitis AI Quantizer using symmetric power-of-two scaling and quantization-aware training with a straight-through estimator, for which they cite TQT [35] (p. 5). They retrain with a tuned quantizer-parameter learning rate and report that the deployed QAT models match the expected GPU performance with little to no degradation (Table 4, Section 4.3, p. 11). TQT is cited by reference number rather than named, and is applied through the AMD/Xilinx Vitis AI toolchain.

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 16](#).

Reviewed version: Published ACM version (open access, CC BY-NC 4.0). **Pinpoint:** Physical PDF p. 5, Section 3.2: INT8 Vitis AI quantization with power-of-two scaling and QAT using a straight-through estimator [35]; p. 11, Section 4.2.2, Table 4 and Section 4.3: QAT with a quantizer parameter learning rate and the deployed FPGA QAT models; bibliography p. 16, reference [35]. Source E-1277; marked copy E-1458.

Count: E-0386, `data.organic_results[11].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 21-SAR-Vessel-MARKED.pdf

Citation coverage across the research portfolio

Entry	Cited work	Profile citations	Records retrieved	Grouped works	Mapped works
01	Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks	227	216	206	146
02	Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware	71	71	69	41
03	Degradation of 2DEG transport properties in GaN-capped AlGaIn/GaN heterostructures at 600° C in oxidizing and inert environments	22	22	20	11
04	Interdigitated Pt-GaN Schottky interfaces for high-temperature soot-particulate sensing	15	15	15	5
05	Training a fully convolutional neural network to route integrated Circuits	13	13	13	6
06	A novel self-consistent model based optimal filter design for the improved dynamic performance of 3-phase PLLs for phase tracking under grid imperfections (Part 2)	4	4	4	3
07	System and method for implementing neural networks in integrated circuits	1	1	1	0
08	Gallium Nitride Microelectronics for High-Temperature Environments	1	1	1	1
09	An ML Compilation Stack for Autonomous Vehicles using MLIR	0	0	0	0
10	Systematic DL Optimizations for Accurate and Efficient TensorRT Deployments on Cruise's Self-Driving Fleet	0	0	0	0
11	TQT: Trained Quantization Thresholds	0	0	0	0

A citing work can appear in several publication rows. Unique survey totals group overlapping records and publication versions.

The saved Scholar profile records 354 citations. The inventory retains 323 works from 337 archived Scholar records. Publication affiliations establish locations for 210 works; the remaining 113 have their review status recorded in the inventory.

Dated Scholar responses document discovery and article-level counts. Source PDFs establish the selected discussions and affiliation evidence. The coverage table and inventory provide the denominators for each measure.

Publication-sourced affiliations and research connections

The map presents 228 institution/location entries from 210 grouped citing works in 30 countries/territories. Each pin opens the institutions, citing papers and affiliation-page evidence. Several institutions in one city share a display pin; multi-institution papers contribute several locations.

Affiliations come from the reviewed publication. Explicit campus and city details determine placement; ROR supplies institution-city geography when the publication lists an institution without a city. Hollow markers show country-level affiliations. Coordinates are geographic display anchors.

The map documents the geographic reach of citing authors' publication affiliations, including coauthor- and employer-linked works. Article passages establish methodological adaptation, implementation, comparison, design connections or technical exposition. Joint attribution is retained where several methods are credited.

Research files

[Marked discussions](#): marked source pages in this report's article order.

Marked-Articles/: complete articles with original page numbers and source links.

Selected-Articles.csv: selection membership, counts, evidence types and source pages.

Discussion-Page-Index.csv: article-to-packet page mapping.

Sources/: dated Scholar, publication, venue, geography and affiliation captures.

[Substantive Citation Report](#) · [Interactive map](#)

Counts refer to the saved October 7, 2026 snapshot. Hashes and source provenance are recorded in the research manifests.